

Human Number Line Race

Number Sequencing and Fun with Numbers 1 to 20

Learning Goals

- **Count:** Count forward from 1 to 20 and backward from 20 to 1 confidently.
- **Order:** Arrange numbers in ascending (small to big) and descending (big to small) order.
- **Sequence:** Identify what number comes before and after a given number.
- **Compare:** Use the concepts of greater than ($>$) and less than ($<$) to compare numbers.
- **Visualise:** Understand the number line as a visual tool for representing numbers.
- **Apply:** Relate number concepts to real-life scenarios like trains, queues, cricket scores, and sweets.



Warm-Up: Rocket Launch Countdown!

Let's get moving and activate our number skills with a fun launch countdown activity:

1. Stand up straight with hands raised high towards the sky.
2. Count down together as a class: 20, 19, 18 ... 3, 2, 1 .
3. On "1", shout "**BLAST OFF!**" and jump up high like an ISRO rocket!
4. Now, let's reverse it: count upwards from 1 to 20 ("1 se 20 tak!").



Meet the Number Line

A **number line** is a straight line where numbers are arranged in a fixed, sequential order. Think of it like train coaches lined up at Chennai Central Railway Station. Just as coaches have definite positions next to each other, every number has a fixed neighbour on either side.

The Three-Coach Model

Before (S6)	Focus Number (S7 Coach)	After (S8)
Coach S6 comes BEFORE S7. In numbers: 6 comes before 7.	Coach S7 is in the MIDDLE . It has neighbours on both sides.	Coach S8 comes AFTER S7. In numbers: 8 comes after 7.

Ordering Numbers: Ascending & Descending

Numbers can move in two directions: going up or going down, just like a staircase.

Ascending Order	Descending Order
Chhote se Bade (Small to Big) Going UP the stairs: 1, 2, 3, 4, 5... <i>Real-life examples:</i> • School assembly: line from shortest to tallest. • Cricket jersey numbers. • Apartment building floors going up.	Bade se Chhote (Big to Small) Going DOWN the stairs: 10, 9, 8, 7... <i>Real-life examples:</i> • ISRO rocket countdown: 10, 9, 8... 0! • Kirana shop stacking tins from heaviest to lightest.

The Staircase Memory Trick

Arrow Direction	Type of Order	Memory Cue	Example
↑ UP	Ascending	Climbing stairs	1, 2, 3, 4, 5
↓ DOWN	Descending	Rocket countdown	10, 9, 8, 7, 6

Before and After Numbers

Understanding positions helps build strong number sense. Here are two everyday ways to remember how numbers sit side by side:

- **The Ration Shop Queue:** People stand in a fixed queue holding token numbers. Token 6 comes **BEFORE** Token 7; Token 8 comes **AFTER** Token 7.

- **Counting Rhyme:**

"Do" (2) comes **AFTER** "Ek" (1)

"Do" (2) comes **BEFORE** "Teen" (3)

Quick Reference

Number Before	Given Number	Number After
4	5	6
9	10	11
14	15	16
19	20	(End of line)
(Start of line)	1	2

Greater Than & Less Than

To compare numbers easily, use the **Crocodile Mouth Trick (Magar Machh)**! The hungry crocodile's mouth **OPENS** towards the **BIGGER** number because it wants a larger meal.

- **Big > Small** (Greater Than)
- **Small < Big** (Less Than)

Comparison Examples

Situation	Numbers	Symbol	How to Read It
Riya has 15 laddoos; Arjun has 9 laddoos	15 and 9	15 > 9	15 is Bada (Greater than) 9
Team A scored 12 runs; Team B scored 17 runs	12 and 17	12 < 17	12 is Chhota (Less than) 17

The Game: Human Number Line Race!

Bring the concepts together in an active team race. Two teams (**Team Mango** and **Team Coconut**) each receive a set of number cards from 1 to 20. Arrange into a living number line according to the rounds:

ROUND 1: Ascending Race

Physically arrange yourselves from smallest to biggest (1, 2, 3 ... 20) along the floor line as fast as possible.

ROUND 2: Descending Race

Reverse your thinking! Quickly re-arrange yourselves from the biggest to the smallest number (20, 19, 18 ... 1).

ROUND 3: Mystery Challenge

When a reference number is called out, look at your card and shout out whether your number comes **BEFORE** or **AFTER** that number!



Quick Summary

- **Number Line (Sankhya Rekha):** Numbers arranged in a fixed sequential order, where every number has a definite position and fixed neighbours.
- **Ascending Order (Chhote se Bade):** Small to big (going UP the stairs, e.g., 1 to 20).
- **Descending Order (Bade se Chhote):** Big to small (going DOWN the stairs, e.g., 20 to 1).
- **Before & After (Pehle aur Baad):** Every number has one number immediately before it and one immediately after.
- **Greater & Less Than (Bada aur Chhota):** The crocodile mouth always opens wide to eat the larger number (15 > 9).



Take-Home Fun

Try these quick games with your family at home to practice your skills:

- **Family Call-Out:** Have a family member call out any number from 1 to 20, and you quickly name the numbers that come before and after it.
- **Card Game:** Draw random playing cards and arrange them in ascending or descending order.

- **Page Race:** Open a book to two different pages and decide which page number is greater using the crocodile rule!

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